COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE 21S1
DATE PERFORMED:	JULY 9 2026	**DATE SUBMITTED:**	JULY 13 2026
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ACTIVITY NO. 1 [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

Section	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	`int getSUM(int first, int last);`
	`void COMPARE(int first, int last);`
	`bool box(bool first, bool last);`

Main Function	`int main() {`
	`std:: cout << getSUM(6,7)<<std::endl;`
	`COMPARE(7,11);`
	`box(true,false);`
	`return 0;`
	`}`

Method Definition	`int getSUM(int first, int last) { return first + last; }`
	`void COMPARE(int first, int last) { if (first > last) { std::cout << first << " is greater than " << last << std::endl; } else if (last > first) { std::cout << last << " is greater than " << first << std::endl; } }`
	`bool box(bool first, bool last) { std::cout << first << " AND " << last << " = " << (first && last) << std ::endl; std::cout << first << " OR " << last << " = " << (first || last) << std ::endl; std::cout << "NOT " << first << " = " << !(first) << std::endl; std::cout << "NOT " << last << " = " << !(first) << std::endl; return true; }`

Table 1-1. C++ Structure Code for Answer

```cpp
#include <iostream>

class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();

};

int main(){
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle.\n";
    }
    else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }
    return 0;
}

Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
void Triangle::setAngles(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}
```

```
> cd "c:\Users\TIPQC\Downloads\temp\HOA_1\" ; if ($?) { g++ ILO_B.cpp -o ILO_B } ; if ($?) { .\ILO_B }
The shape is a valid triangle.
PS C:\Users\TIPQC\Downloads\temp\HOA_1>
```

With the applications of private and public syntaxes for class, It brings up new possibilities that I've never actually explored yet. One of my observations is that the private variables are just exclusive to the class parent, which makes it more organized and minimal. I would like to comment that I myself tend to hard code when it gets complicated, but this method is much more simpler and unexpectedly easier to understand.

Table 1-2. ILO B output observations and comments.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Create a C++ program to swap the two numbers in different variables

```cpp
#include <iostream>

int main() {
    int a = 2;
    int b = 3;
    int c;
    std::cout << a << " " << b << std::endl;
    c = a; a = b; b = c;
    std::cout << a << " " << b << std::endl;
}
```

```
PS C:\Users\TIPQC\Downloads\temp\HOA_1> cd "c:\Users\TIPQC\Downloads\temp\HOA_1\" ; if ($?) { g++ SUP_ACT1.cpp -o SUP_ACT1 } ; if ($?) { .\SUP_ACT1 }
2 3
3 2
PS C:\Users\TIPQC\Downloads\temp\HOA_1>
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
#include <iostream>

double KtoF(double k);

int main(){
    std::cout << KtoF(300);
}

double KtoF(double k) {
    return (((k - 273.15)*(1.8))+ 32);
}
```

```
> cd "c:\Users\TIPQC\Downloads\temp\HOA_1\" ; if ($?) { g++ SUP_ACT2.cpp -o SUP_ACT2 } ; if ($?) { .\SUP_ACT2 }
80.33
PS C:\Users\TIPQC\Downloads\temp\HOA_1>
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
#include <iostream>

int DIST(int first, int end);

int main() {
    std::cout << DIST(23,98);
}

int DIST(int first, int end) {
    return end - first;
}
```

```
PS C:\Users\TIPQC\Downloads\temp\HOA_1> cd "c:\Users\TIPQC\Downloads\temp\HOA_1\" ; if ($?) { g++ SUP_ACT3.cpp -o SUP_ACT3 } ; if ($?) { .\SUP_ACT3 }
75
PS C:\Users\TIPQC\Downloads\temp\HOA_1>
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```
...OA_1 > SUP_AC14.cpp > ...
#include <iostream>

class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
        void typeT(double A, double B, double C);
};

float areaT(float b, float h);
float periT(float a, float b, float c);

int main(){
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle.\n";
    }
    else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }
    std::cout << "Area: " << areaT(20,10) << std::endl;
    std::cout << "Perimeter: " << periT(6,7,8) << std::endl;
    Triangle angles(0,0,0);
    angles.typeT(30,40,10);
    return 0;
}

Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
void Triangle::setAngles(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}
```

```
46  void Triangle::typeT(double A, double B, double C){
47      if ( C*C < A*A + B*B) {
48          std::cout << "Type: Acute Triangle";
49      }
50      else if (C*C == A*A + B*B) {
51          std::cout << "Type: Right Triangle";
52      }
53      else if (C*C > A*A + B*B) {
54          std::cout << "Type: Obstuse Triangle";
55      }
56      else {
57          std::cout << "Type: Others";
58      }
59  }
60  float areaT(float b, float h) {
61      return 0.5 * (b * h);
62  }
63  float periT(float a, float b, float c) {
64      return a + b + c;
65  }
```

```
"D:\-COLLEGE\SECOND YEAR\CODING\CLION EXCLUSIVE\cmake-build-debug\CLION_EXCLUSIVE.exe"
The shape is a valid triangle.
Area: 100
Perimeter: 21
Type: Acute Triangle
Process finished with exit code 0
```

*performed using CLION at home

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of Concepts and Information concerning Geometry**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was only used to explain information such as area, perimeter and the type of triangle.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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